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Your Roll No. 20BCS021

CEN-503

B.Tech.(Computer Engineering) V Semester Examination 2022

Microprocessor

Paper No. : CEN-503

Time: 3 Hours

Maximum Marks: 45

(Write your Roll No. on top immediately on receipt of this question Paper)

Note: - Attempt all questions.

CO1

Q1. (a). Draw and explain functional block diagram of 8085 microprocessor. 4.5

(b). Discuss how microprocessor 8086 is different from microprocessor 8085. 4.5

OR

Q1'(a) Draw and explain timing diagram of SHLD 2060 H instruction of 8085 stored at address 2500H. 4.5

Q1'(b)Draw and Discuss Pin diagram of 8085 Microprocessor 4.5

CO2

2(a).

(i) Explain the function of the following program 1.5

```
      MVI  A , Byte
      ORA  A
      JM   NEXT
      OUT 80 H
       HLT
NEXT: CMA
      ADI 01H
      OUT 80H
      HLT
```

CO3

(ii) What is the output at PORT1 when the following instructions are executed? Explain. 1.5

```
        MVI  A , 9FH
        ADI  82H
        JC   Display
        OUT  PORT1
        HLT
Display: XRA  A
        OUT  PORT1
        HLT
```

(iii) In the following program explain the range of the bytes that will be displayed at PORT2 1.5

```
        MVI  A , BYTE1
        MOV B, A
        SUI  60H
        JC   DELETE
        MOV A , B
        SUI  90H
        JC DISPLAY
DELETE :  XRA A
        OUT  PORT1
        HLT
DISPLAY : MOV  A , B
        OUT  PORT2
        HLT
```

2(b) Explain the meanings of the following instructions of 8085/8086 after writing their syntax. Also illustrate their use with the help of suitable example. 4.5

(i)STAX

(ii)SHLD

(iii)LOOP

(iv)MOVSB

CO3

3(a). Write a well documented program in assembly language of 8085 for 8X8 multiplication without shifting multiplicand. 4.5

3(b). Write a well documented program in the assembly language of 8085 for 16x8 division. 4.5

OR

3'(a).Write a program in assembly language of 8085 to arrange 400 integers stored at consecutive memory locations starting with address 2300H in Ascending order. Program should be well documented. 4.5

3'(b)Write a program in assembly language of 8085 to generate Modulo-10 counter with a delay of 90 ms between each count assuming a system with 0.5 microsecond clock period. Show all calculations. 4.5

CO4

4(a). Explain how I/O devices are interfaced as Memory Mapped I/O with help of suitable diagram and examples. 4.5

4(b). What is PPI ? How are peripheral connected to microprocessor through 8255? Explain with the help of suitable example and diagrams. 4.5

CO5

5. (a). Write a program in assembly language of 8086 to find out the number positive and no of negative numbers out of N 16-bit numbers stored at consecutive memory locations. 4.5

5(b). Discuss how assembly language is used with C programming with suitable example. Also discuss how are software interrupt used in programming with suitable example. 4.5